



Rajeev Mutyalu

AI Workflows Architect • Creative Technology Lead • VFX & GenAI Systems Architect

☆ GenAI Production Systems • Creative Workflows • 1917 • RRR • Mufasa

● Former Technicolor R&D Supervisor • Technicolor Fellowship Network Associate Member

UK Skilled Worker Visa • Existing Sponsorship in Place • Available Immediately

LOCATION: London, United Kingdom

PHONE: +44 7827 498399

EMAIL: mutyalu.rajeev@gmail.com

PORTFOLIO: Portfolio ↗

LINKEDIN: linkedin.com/in/rajeevmuthyalu

GITHUB: github.com/rajeev-mutyalu

Executive Summary

AI Workflows Architect, Creative Technology Lead and former Technicolor R&D Supervisor with 20+ years of experience across VFX production technology, software architecture, creative engineering and global engineering leadership. Currently architecting Generative AI production systems at Astra Studios, including the **Studio.AI platform** integrating **Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node workflows** with character identity consistency, artist-facing controls and ACES colour conforming.

Combines emerging generative video technology with established filmmaking workflows, DCC APIs including Nuke, Maya and Hiero, production-grade OpenUSD systems, event-driven automation and private AI infrastructure. Proven track record of taking emerging technologies from R&D into production, including ReviewTool, ProNuke, Meshroom photogrammetry, Editorial Turnover / Conform systems and modern AI-enabled production workflows across major productions including [1917\(Academy Award for Best Visual Effects\)](#), [RRR](#), [Mufasa: The Lion King](#), [Back in Action](#), [Spaceman](#), and [Prehistoric Planet](#). Associate Member of the Technicolor Fellowship Network Committee and recipient of 9+ studio leadership & excellence awards.

Architecture & Technical Competencies

Generative Video

Generative AI Platforms & Workflows

- Generative AI Platforms:** Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node workflows, latent consistency models.
- Productionisation:** Studio.AI platform, controlled generation workflows supporting character and visual continuity, artist-facing controls and ACES colour conforming.

Leadership

Engineering Leadership & Delivery

- Team & Technical Leadership:** Leading R&D software engineering, development, and TD teams across London, Montreal, and Bengaluru.
- Production Execution:** Translating creative requirements into technical systems, taking R&D technologies into production, Agile/Scrum delivery.

Agentic & MCP

Agentic AI & MCP Systems

- Model Context Protocol:** Custom MCP server engineering, JSON-RPC 2.0, tool/resource schemas and pipeline API binding.
- Agentic Workflows:** Claude Code, OpenAI Codex, Google Antigravity, subagent orchestration, task decomposition and multi-step tool workflows.

OpenUSD

OpenUSD Production Architecture

- USD Architecture:** OpenUSD (USD Python API, composition arcs, layer stacking, edit targets, sublayering), full-CG & live-action shot pipelines.
- Colour & Timeline Standards:** OpenColorIO (OCIO), ACES colour pipeline, OpenTimelineIO (OTIO) sequence interchange.

Core Systems

Core Software & Systems Architecture

- Languages & UI:** Python 3.x (Expert), PyQt / PySide (UI/UX engineering), C++, Java, Objective-C, Cocoa, shell scripting.
- Infrastructure:** Docker, Linux/Unix environments, REST APIs, FastAPI, Git, Tractor render farm management & dispatching.

AI Architecture

AI Production Architecture

- AI Production Architecture:** Multi-model workflow integration, artist-facing controls, structured prompt routing, production ingest and editorial integration.
- Creative Production Systems:** Connecting generative AI outputs with established VFX, review, colour and production workflows.

Production Systems

VFX Production Systems & DCC APIs

- Review & Assembly:** ReviewTool (real-time 4K+ media caching & multi-facility synchronisation), ProNuke modular shot assembly, Meshroom photogrammetry.
- DCC & Studio APIs:** Nuke C++/Python API, Maya Python API, Hiero / Nuke Studio Python API, ShotGrid Python API, 3D Equalizer, Silhouette.

Private AI

Private / On-Premise AI

- Local Inference:** Ollama, air-gapped studio inference, local private LLM deployments, GGUF / AWQ quantisation, context-window optimisation.
- Secure Endpoints:** Secure internal AI endpoints designed to keep sensitive studio data within controlled infrastructure and reduce exposure to public AI services.

Conform & I/O

Editorial Turnover, Conform & I/O

- Conform & Turnover:** Package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero conform.
- Grade & Asset Processing:** CDL grades, shot LUT processing, frame range validation, USD plate sublayering.

Automation

Production Automation & Integration

- n8n Workflow Hub:** Inbound webhooks, reverse webhooks, asynchronous event routing, production API synchronisation.
- Event Automation:** Automated show onboarding, smart plate validation, automated ShotGrid metadata injection.

Professional Experience

CURRENT LEADERSHIP ROLE

Sep 2025 - Present

Lead Software Architect

Astra Studios • London, UK

Focus: Generative AI, Creative Systems & Production Technology

- **Engineering & Technical Leadership:** Lead Development and TD teams, defining production technology architecture and engineering standards while working directly with artists, production, supervisors and leadership to translate creative requirements into scalable technical systems.
- **Studio AI GenAI Platform & Video Workflows:** Engineered the Studio AI generative platform from architecture to production, integrating prompt-to-video engines (Google Veo, Kling AI, Higgsfield, Seedance and ComfyUI node workflows) with character identity consistency, artist-facing controls, and ACES colour conforming.
- **Production VFX & OpenUSD Pipeline:** Architected an end-to-end production-grade OpenUSD pipeline bridging DCCs (Maya, Nuke) from modelling to compositing for full-CG and live-action VFX.
- **Editorial Turnover, Conform & I/O:** Engineered the studio's Editorial Turnover, Conform & I/O pipeline from scratch, integrating turnover package inspection, automated DI discrepancy reporting, regex pattern generation, multi-track Hiero conform, CDL grades, shot LUTs, and USD plate sublayering.
- **Zero-Touch Studio Automation:** Orchestrated zero-touch n8n studio automation hubs using inbound and reverse webhooks for automated show onboarding, smart plate validation, and ShotGrid metadata injection.

R&D Supervisor

Mar 2023 - Feb 2025

Technicolor Group • London, UK

- **Global Review Platform Architecture:** Spearheaded the architecture and productionisation of ReviewTool, a studio-wide VFX review platform with real-time 4K+ media caching, multi-facility synchronisation and RV integration supporting distributed production workflows across global studios.
- **ProNuke Modular Ecosystem:** Engineered ProNuke modular shot assembly with Gather UI for automated shot builds, Release UI for sequence-level publishing, and Batch Manager bulk render automation across global artist teams.
- **Meshroom Photogrammetry in Production:** Productionised Meshroom into matchmove and tracking workflows, cutting camera extraction turnaround from days to hours with fast switching between Meshroom, Maya, and 3D Equalizer.
- **Global R&D Leadership:** Led globally distributed R&D software engineering teams across London, Montreal, and Bengaluru supporting Compositing, Roto, Paint Prep, Camera Tracking, and Rotoanim departments.
- **Conform Ingest Automation:** Supervised Conform Ingest workflows using Hiero and Nuke Studio, automated CDL and LUT grade processing, and ShotGrid frame range validation.

R&D Supervisor

Jan 2022 - Feb 2023

Technicolor Group • Bengaluru, India

- **2D Software Engineering:** Headed 2D software engineering and pipeline development across Bengaluru and Montreal hubs.
- **Review Platform Prototyping:** Architected core compositing automation tools and early prototypes of the studio VFX review platform (ReviewTool).
- **Agile Execution:** Mentored engineers, established Agile and Scrum sprint frameworks, and aligned dev priorities with production executives.

Lead Pipeline Software Developer

Jul 2017 - Dec 2021

MPC Film • Bengaluru, India

- **Landmark Production Pipelines:** Led core 2D pipeline solutions across Academy Award-winning films and global tentpole productions (1917, RRR, The Lion King).
- **Cross-Studio Platform Standardisation:** Spearheaded RPM Unification across MPC, MR.X and The Mill, consolidating and standardising software toolsets and UI/UX across compositing and tracking workflows to improve technical consistency across multiple studio environments.
- **Nuke Bridge Hub:** Developed Nuke Bridge integrating Tessa (AMS), ShotGrid (PMS), and Tractor (Render Farm), establishing a version-controlled, high-throughput pipeline.
- **Render Dispatcher:** Engineered the Nuke Render Dispatcher with automated render order traversal and Reserve & Commit dispatch mechanics.

Team Lead – Technology & R&D Production

Jul 2012 - Jun 2017

Technicolor • Bengaluru, India

- **Media Engines:** Architected macOS AVFoundation and QuickTime media processing engines (Channel Mapper, Frame Rate Converter, Compliance Edit Lists).
- **Team Management:** Managed an engineering team of 7 developers across C++, Objective-C, Cocoa, and Java, optimising production compression and authoring pipelines.
- **Hardware Compliance:** Served as primary technical contact for player manufacturers, resolving Blu-ray authoring framework and playback compatibility issues.

Senior Software Engineer – Technology

May 2008 - Jun 2012

Technicolor • Bengaluru, India • Los Angeles, USA • Hannover, Germany

- **Disney Collaboration (Los Angeles):** Collaborated on-site with Disney engineering in Los Angeles to architect the Fast Menu Load (FML) template and integrate Generation 2 Bootloaders (G2B).
- **World First 3D Blu-ray (Hannover):** Engineered core 3D capabilities into the BD-J Authoring framework at Technicolor R&I in Hannover, Germany, as a key contributor to the world's first released 3D Blu-ray disc.
- **Founding Engineer:** Founding software engineer at Technicolor India, building runtime alignment engines ('Graphics Adjuster') and studio authoring templates.

Application Engineer & Game Programmer

Sep 2006 - Apr 2008

e4e & Dhruva Interactive • Bengaluru, India

- Ported mobile game titles to Nokia, Motorola, Samsung, and Sony Ericsson devices for T-Mobile and Verizon; engineered touchscreen interfaces for early mobile OS platforms.

Honours, Awards & Fellowship

**Technicolor Fellowship Network**

2015 - 2025

Associate Member • Technicolor Group

“Awarded for production environment expertise, strong customer relationships, and deep collaboration with Research & Innovation—delivering a rare mix of creativity, pragmatism, and architectural stewardship.”

— Fellowship Nomination Committee • Mar 2015

Fellowship Committee Election R&I Collaboration

Architectural Leadership

**Studio Leadership & Excellence**

2008 - 2024

9+ Honours • Technicolor, MPC Film & e4e

 **Value Champions & Best Employee of the Month (Aug 2019)** — Technicolor FEV

 **Performer of the Quarter • Team Lead (Oct 2016)** — Technology R&D leadership

 **Performer of the Quarter (2011, 2012, 2014)** — Disney collaboration & pipeline innovation

 **Extra Miler & Executive Team Awards (2011, 2013)** — DVD/Blu-ray software

Recent AI Professional Development

AI Generalist Accelerator Program • Generative AI Business Applications, Outskill (Aug 2026)

[View Certificate ↗]

Workflow automation (n8n), custom Model Context Protocol (MCP) tooling, agentic architectures, and prompt engineering.

AI Accelerator C13 Hackathon • 2-Day Innovation Challenge, Outskill (Aug 2026)

[View Award ↗]

Rapid prototyping, agent system architecture, and production deployment of generative AI application workflows.

Education & Academic Background

Bachelor of Technology (B.Tech) • Computer Science & Engineering

2002 - 2006

Jawaharlal Nehru Technological University (JNTU) • Sri Venkateswara College of Engineering & Technology (SVCET)